# Agorithm and Programming Practicum Report

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**TEKNIK INFORMATIKA**

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**SINGARAJA**

A. Introduction

The Fibonacci sequence is a sequence of numbers where each element is the sum of the two previous elements, starting from 0 and 1. This web will implement the Fibonacci sequence using recursion and iteration. In addition, it will also calculate the volume of various spatial figures, such as cubes, blocks, tubes, and spheres. I will use appropriate mathematical formulas to calculate the volume of each of these shapes.

B. Javascript Code Explanation

```
const fibonacciButton = document.getElementById('fibonacciButton');
Comment Code
fibonacciButton.addEventListener('click', function () {
    let n = parseInt(document.getElementById('insertFibonacci').value);
    let resultFibonacci = document.getElementById('resultFibonacci');

    if (n <= 0) {
        resultFibonacci.textContent = "The n must be bigger than 0";
    } else {
        let a = 0;
        let b = 1;
        let result = 0;

        for (let i = 2; i <= n; i++) {
            result = a + b;
            a = b;
            b = result;
        }

        resultFibonacci.textContent = "The " + n + " Fibonacci sequence is :" + result;
    }
});
```

When the "fibonacciButton" is clicked, the programme will take the user input (n), check if n is greater than 0, and then calculate the Fibonacci sequence up to the order of n. The result is then displayed on the HTML element "resultFibonacci". The result is then displayed on the HTML element with the ID "resultFibonacci".

```
//KUBUS
const cubeButton = document.getElementById('cubeButton');

Comment Code
cubeButton.addEventListener('click', function (){
    const cubeSide = document.getElementById('cubeSide').value;

    const volume = Math.pow(cubeSide, 3);
    const formattedVolume = volume.toFixed(2);

    document.getElementById('resultCube').textContent = 'Result: ' + formattedVolume;
});
```

The JavaScript code above aims to calculate the volume of a cube. When the user clicks on the button with ID "cubeButton", the programme retrieves the side value of the cube from the HTML element with ID "cubeSide", then calculates the volume of the cube by lifting the side

value to the power of 3. The result is formatted to two decimals and displayed in the HTML element with ID "resultCube".

```
cuboidButton.addEventListener('click', function () {
    const length = parseFloat(document.getElementById('cuboidLong').value);
    const width = parseFloat(document.getElementById('cuboidWidth').value);
    const height = parseFloat(document.getElementById('cuboidHeight').value);


    const volume = length * width * height;
    const formattedVolume = volume.toFixed(2);

    document.getElementById('resultCuboid').textContent = 'Result: ' + formattedVolume;
});
```

The JavaScript code above aims to calculate the volume of a block (cuboid). When the user clicks the button with the ID "cuboidButton", the programme retrieves the length, width, and height values from the corresponding HTML element. Next, the programme calculates the volume of the block by multiplying the three dimensions, formats the result to two decimals, and displays the volume result on the HTML element with ID "resultCuboid".

```
const ballButton = document.getElementById('ballButton');

Comment Code
ballButton.addEventListener('click', function () {
    const radius = parseFloat(document.getElementById('radiusBall').value);

    if (isNaN(radius) || radius < 0) {
        alert('Please enter a valid positive number for the ball radius.');
        return;
    }

    const volume = (4/3) * Math.PI * Math.pow(radius, 3);

    const formattedVolume = volume.toFixed(2);

    document.getElementById('resultBall').textContent = 'Result: ' + formattedVolume;
});
```

This JavaScript code is used to calculate the volume of the ball based on the radius entered by the user. When the button with ID "ballButton" is clicked, the programme retrieves the radius value, checks if the value is a valid positive number, and if it is, calculates the volume of the ball using the appropriate mathematical formula. The volume result is formatted with two decimals and displayed on the web page in the element with ID "resultBall". If the entered radius is invalid, the programme will provide a warning message to the user.

```
const cylinderButton = document.getElementById('cylinderButton');

Comment Code
cylinderButton.addEventListener('click', function () {
    const radius = parseFloat(document.getElementById('cylinderRadius').value);
    const height = parseFloat(document.getElementById('cylinderHeight').value);

    if (isNaN(radius) || isNaN(height) || radius < 0 || height < 0) {
        alert('Please enter valid positive numbers for the cylinder radius and height.');
        return;
    }

    const volume = Math.PI * Math.pow(radius, 2) * height;

    const formattedVolume = volume.toFixed(2);

    document.getElementById('resultCylinder').textContent = 'Result: ' + formattedVolume;
});
```

The JavaScript code above is used to calculate the volume of a cylinder. When the user clicks on the button with the ID "cylinderButton", the programme retrieves the radius and height values of the cylinder from the corresponding HTML element. Then, the programme checks whether the values are valid positive numbers. If they are valid, the program calculates the volume of the cylinder using the appropriate mathematical formula, formats the result to two decimals, and displays it on the web page in the element with ID "resultCylinder".

```
//KERUCUT
const coneButton = document.getElementById('coneButton');

Comment Code
coneButton.addEventListener('click', function () {
    const radius = parseFloat(document.getElementById('radiusCone').value);
    const height = parseFloat(document.getElementById('coneHeight').value);

    if (isNaN(radius) || isNaN(height) || radius < 0 || height < 0) {
        alert('Please enter valid positive numbers for the cone radius and height.');
        return;
    }

    const volume = (1/3) * Math.PI * Math.pow(radius, 2) * height;

    document.getElementById('resultCone').textContent = 'Result: ' + volume.toFixed(2);
});
```

The JavaScript code above is used to calculate the volume of a cone. When the user clicks the button with the ID "coneButton", the programme retrieves the radius and height values of the cone from the corresponding HTML element. Then, the programme checks whether the values are valid positive numbers. If they are, the program calculates the volume of the cone using the appropriate mathematical formula, formats the result to two decimals, and displays it on the web page in the element with ID "resultCone".

C.Github Page

https://gabrielnathanael.github.io/

D. GitHub Repository

https://github.com/GabrielNathanael/GabrielNathanael.github.io